

OmegaSim/OmegaHive Strange-Attractor Tuning: A Hyperseed-Dynamical Formalization

OmegaClaw / ZeroBot working notes for Ben Goertzel

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Abstract

This note formalizes a useful mathematical frame for constructing and tuning OmegaSim/OmegaHive-style simulations so that they can explore complex bounded nonlinear dynamics, including operational strange-attractor regimes. The core idea is to model hives as delayed, coupled, dissipative nonlinear systems; to express tuning through dimensionless control ratios; to prove what follows from well-posedness, boundedness, contraction, and delay-instability arguments; and to translate the resulting theory into concrete simulation recommendations. The claims here are deliberately split into theorems, conjectures, operational criteria, and implementation suggestions.

1 Provenance and aim

This note responds to Ben Goertzel’s 2026-06-28 request for a rigorous mathematical formalization of the “useful frame” proposed for tuning one-hive and OmegaHive-style simulations toward complex strange-attractor dynamics. The requested steps are:

1. formalize concepts bridging Hyperseed and nonlinear dynamics;
2. state and prove general-purpose theorems applicable across agent hives;
3. specialize these results to OmegaHive1/OmegaSim-style simulations;
4. give theory-motivated instructions for the Codex/OmegaSim simulation agent.

The purpose is not to prove that a given current simulation has a strange attractor. Rather, it is to identify mathematically meaningful design knobs and verification gates so that such a claim can later become evidence-bearing.

2 S1. Concepts bridging Hyperseed and nonlinear dynamics

2.1 Hyperseed event schema as a state observable

Definition 1 (Hyperseed process event). *A Hyperseed-relevant process event is a typed record*

$$e = \langle \iota, \tau, A, C, R, P, \Theta, \nu \rangle$$

where ι is identity, τ time, A the active agents or sub-processes, C context, R realized relation or result, P provenance, Θ an interpretive type assignment, and ν uncertainty/truth-value metadata.

Definition 2 (Hive observable). *A hive observable is a map*

$$O : E_{\leq t} \rightarrow X \subseteq \mathbb{R}^n$$

from a finite or filtered history of Hyperseed process events to a numerical state vector. Components may include agent activation, resource load, attention, coordination phase, memory pressure, trust, artifact quality, task backlog, message volume, or semantic diversity.

Definition 3 (Scrutable reduction). *A reduction from Hyperseed events to dynamical state is scrutable when each coordinate x_i of $x = O(E)$ has: (i) a declared meaning, (ii) a measurement rule, (iii) provenance to events or logs, and (iv) an uncertainty model. Unscrutable coordinates may be useful for machine learning, but should not carry ontological interpretation without an additional explanation layer.*

2.2 Delayed agent-hive dynamics

Let $x(t) \in X \subseteq \mathbb{R}^n$ be a numerical hive state derived from the observable map above. Let the delayed history segment be

$$x_t(\theta) = x(t + \theta), \quad \theta \in [-\tau, 0],$$

where $\tau \geq 0$ is a maximum communication, reaction, or memory delay.

Definition 4 (Continuous delayed hive system). *A continuous delayed hive system is a retarded functional differential equation*

$$\dot{x}(t) = F(x_t, u(t), \eta(t); \theta),$$

where $F : C([-\tau, 0], X) \times U \times E \rightarrow \mathbb{R}^n$, $u(t)$ is a controllable intervention/tuning signal, $\eta(t)$ is bounded environmental or stochastic forcing, and θ denotes fixed parameters.

Definition 5 (Discrete delayed hive simulation). *A discrete delayed hive simulation is a lifted map*

$$x_{k+1} = G(x_k, x_{k-1}, \dots, x_{k-m}, u_k, \eta_k; \theta),$$

with lifted state

$$z_k = (x_k, x_{k-1}, \dots, x_{k-m}) \in X^{m+1}.$$

Equivalently,

$$z_{k+1} = \hat{G}(z_k, u_k, \eta_k; \theta) = (G(x_k, \dots, x_{k-m}, u_k, \eta_k), x_k, \dots, x_{k-m+1}).$$

2.3 Dimensionless tuning coordinates

The raw constants of a hive simulation are usually too numerous to sweep intelligently. We therefore define dimensionless ratios.

Definition 6 (Dimensionless hive control ratios). *For a delayed nonlinear hive model, define:*

$$\rho = g r(A), \quad \delta = \frac{\tau}{T_{\text{relax}}}, \quad \mu = \frac{T_{\text{memory}}}{T_{\text{decision}}},$$

$$\kappa = \frac{\|A_{\text{cross}}\|}{\|A_{\text{within}}\|}, \quad \nu = \frac{\sigma_\eta}{\|F_0\| + \epsilon},$$

where g is a typical nonlinear response gain, $r(A)$ is the spectral radius of an interaction matrix or linearized coupling operator, T_{relax} is local relaxation time, T_{memory} memory decay time, T_{decision} update time, A_{cross} and A_{within} are cross-hive and within-hive couplings, and σ_η is noise magnitude.

Interpretation: ρ measures feedback amplification, δ delay pressure, μ persistence, κ modular leakage, and ν noise-to-drift. These are not universal constants; they are organizing coordinates for phase diagrams.

Definition 7 (Operational strange-attractor regime). *A parameter region $\mathcal{P}$ is an operational strange-attractor regime for a simulation if, for a specified observable O and lifted state z , the following hold robustly over validation runs:*

1. *trajectories are ultimately bounded and not merely saturating at hard numerical limits;*
2. *the largest finite-time Lyapunov estimate is positive under controlled perturbations of initial histories;*
3. *long-run statistics are stable under timestep/refinement and seed checks;*
4. *recurrence plots, broadband spectra, or reconstructed dimensions indicate non-periodic structured recurrence rather than white noise.*

This is an empirical criterion. It is weaker than a proof of a mathematical strange attractor.

3 S2. General theorems

3.1 Well-posedness and boundedness

Theorem 1 (Local well-posedness for delayed hives). *Assume F is locally Lipschitz in the history variable x_t , uniformly over bounded u and η . Then for every continuous initial history $\phi \in C([- \tau, 0], X)$ there exists a unique maximal solution.*

Proof. This is the Picard–Lindelof theorem for retarded functional differential equations. The history space $C([- \tau, 0], X)$ is a Banach space under the sup norm. Local Lipschitz continuity gives a contraction on a sufficiently short interval. Repetition extends the solution until it exits every bounded subset of the state space. $\square$

Theorem 2 (Ultimate boundedness from dissipativity). *Suppose there is a radially unbounded C^1 function $V : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ and constants $a, b > 0$ such that along all trajectories with bounded input magnitude M ,*

$$\dot{V}(x(t)) \leq -aV(x(t)) + bM.$$

Then

$$\limsup_{t \rightarrow \infty} V(x(t)) \leq \frac{bM}{a}.$$

Proof. By Gronwall’s inequality,

$$V(t) \leq e^{-at}V(0) + \frac{bM}{a}(1 - e^{-at}).$$

Taking $t \rightarrow \infty$ gives the result. $\square$

Corollary 1 (Attractor-talk requires a bounded regime). *A simulation run that diverges, explodes numerically, or only bounces against uninterpreted clipping bounds does not by itself justify talk of an attractor. Dissipativity or empirically verified boundedness is a prerequisite diagnostic.*

Theorem 3 (Forward-invariant box for delayed maps). *For*

$$x_{k+1} = G(x_k, \dots, x_{k-m}, u_k),$$

assume there is a compact set $K \subseteq \mathbb{R}^n$ such that

$$G(K^{m+1}, U) \subseteq K.$$

Then K^{m+1} is forward invariant under the lifted map $\hat{G}$.

Proof. If $z_k = (x_k, \dots, x_{k-m}) \in K^{m+1}$, then each delayed component lies in K . By assumption the new component $x_{k+1} = G(\dots)$ lies in K . The remaining components of z_{k+1} are shifted copies of previous components in K . Therefore $z_{k+1} \in K^{m+1}$. $\square$

3.2 Contraction: when complex dynamics is impossible

Theorem 4 (Delayed-map contraction implies a unique attracting history). *Let $\hat{G} : K^{m+1} \rightarrow K^{m+1}$ be the lifted delayed map on compact convex K^{m+1} , and suppose it is a contraction with Lipschitz constant $L < 1$ under some norm. Then there is a unique fixed lifted state z^* , and every trajectory converges to z^* .*

Proof. This is Banach's fixed-point theorem applied to $\hat{G}$ on the complete metric space K^{m+1} . Iterates of a contraction converge to the unique fixed point. $\square$

Corollary 2 (Low-gain hives cannot be chaotic). *If the effective lifted update of a hive simulation is contractive, then it cannot have a strange attractor, quasiperiodic orbit, or sustained multi-cycle behavior. Therefore one route toward complex dynamics is to increase gain, delay, cross-coupling, or memory persistence until the lifted system leaves the contraction regime while preserving boundedness.*

Proposition 1 (A practical Lipschitz bound). *Suppose*

$$G(x_k, \dots, x_{k-m}) = S \left(B + \sum_{j=0}^m A_j x_{k-j} \right)$$

where S is componentwise Lipschitz with constant s and the lifted norm is chosen so that the shift has unit weight. A sufficient condition for contraction is that the induced lifted Lipschitz constant of the companion map is less than one; in particular the response gain s and delayed coupling norms $\|A_j\|$ provide an upper bound. Thus the product “gain times coupling” is the first order control axis.

Proof. The non-shift component changes by at most $s \sum_j \|A_j\| \|\Delta x_{k-j}\|$ under the chosen induced norm, while the shift components copy previous differences. Weighted norms can penalize older delay coordinates so that the whole companion map is contractive when the response and coupling norms are sufficiently small. This proves the sufficient, not necessary, condition. $\square$

3.3 Delay-induced oscillatory instability

Theorem 5 (Scalar delay Hopf boundary). *Consider*

$$\dot{x}(t) = -ax(t) + bx(t - \tau), \quad a > 0.$$

The characteristic equation is

$$\lambda + a - be^{-\lambda\tau} = 0.$$

A Hopf crossing candidate satisfies $\lambda = i\omega$, hence

$$a = b \cos(\omega\tau), \quad \omega = -b \sin(\omega\tau).$$

Therefore delayed feedback can destabilize an equilibrium through an oscillatory root crossing even when the local undelayed term is dissipative.

Proof. Substitute $x(t) = e^{\lambda t}$ to obtain the characteristic equation. At an oscillatory stability boundary $\lambda = i\omega$. Separating real and imaginary parts gives the two displayed equations. $\square$

Corollary 3 (Why $\delta = \tau/T_{\text{relax}}$ matters). *The ratio of communication/memory delay to local relaxation time is a genuine bifurcation coordinate, not just an implementation detail. Increasing delay can convert damped consensus into oscillation and, in higher-dimensional nonlinear systems, into period-doubling or more complex regimes.*

3.4 Existence of compact attractors

Theorem 6 (Compact global attractor under dissipative smoothing). *For a continuous delayed hive semiflow on $C([-\tau, 0], X)$, assume: (i) unique forward solutions exist, (ii) solutions enter a bounded absorbing set, and (iii) the semiflow is asymptotically compact. Then there is a compact global attractor $\mathcal{A}$ attracting all bounded sets.*

Proof. This is the standard global-attractor theorem for continuous semiflows: a bounded absorbing set plus asymptotic compactness implies the omega-limit of the absorbing set is nonempty, compact, invariant, and attracts every bounded set. $\square$

Corollary 4 (Delay makes rigor harder). *Delayed systems are infinite-dimensional in history space. Compactness and low fractal dimension are not automatic. OmegaSim should therefore treat finite state-history simulations as approximations and validate attractor claims under history-length and timestep variation.*

3.5 What remains conjectural

Conjecture 1 (Useful complex regime window). *For many OmegaHive-like systems there is a practical window between contraction and blow-up/saturation, characterized roughly by $\rho \approx 1$, non-negligible δ , moderate μ , and controlled κ , in which bounded complex recurrence is most likely.*

This is not a theorem because it depends on nonlinear form, topology, stochastic forcing, and measurement. It is a theory-guided search heuristic.

Conjecture 2 (Hyperseed-scrutable complexity). *The most scientifically useful OmegaSim regimes will be those where the low-dimensional observables show complex recurrence while Hyperseed event records remain interpretable enough to explain which agent roles, memories, artifacts, and feedback loops generate the recurrence.*

4 S3. Specialization to OmegaHive1 / OmegaSim

Since the exact OmegaHive1 simulation state is still evolving, define a generic OmegaHive1 state vector as

$$x_k = (a_k, r_k, m_k, q_k, h_k, c_k, p_k) \in \mathbb{R}^n,$$

where a is agent activation/attention, r resource or compute load, m memory persistence/pressure, q artifact quality or progress, h semantic diversity, c coordination/coherence, and p task pressure. The components may be vectors indexed by agents, roles, artifacts, or sub-hives.

A useful discrete OmegaHive1 skeleton is

$$x_{k+1} = S(W_0 x_k + W_1 x_{k-d} + W_m M_k + B u_k + \xi_k) - D x_k,$$

with memory update

$$M_{k+1} = (1 - \alpha) M_k + \alpha H(x_k, \text{events}_k),$$

where S is saturating nonlinear response, W_0 present-time coupling, W_1 delayed coupling, M_k memory state, u_k interventions, ξ_k controlled noise, and D local decay.

Proposition 2 (OmegaHive1 boundedness by saturation and resource accounting). *If each component of S maps into a compact interval K_i and resource/memory updates are either projected into compact budgets or dissipative in the sense of the boundedness theorem, then the lifted OmegaHive1 history has a compact forward invariant set.*

Proof. The saturating components satisfy the invariant-box theorem. Resource and memory coordinates either remain in explicit compact budgets or satisfy dissipative bounds. The product of these compact coordinate sets is compact and invariant under the lifted delayed map. $\square$

Proposition 3 (OmegaHive1 contraction baseline). *If the fitted or designed Lipschitz constant of the lifted OmegaHive1 update is below one, the system will converge to a unique stable regime and should not be expected to exhibit strange-attractor-like dynamics.*

Proof. Immediate from the delayed-map contraction theorem. $\square$

Proposition 4 (OmegaHive1 delay/gain destabilization target). *Let x^* be an equilibrium of a smooth deterministic OmegaHive1 skeleton. Linearizing gives*

$$\delta x_{k+1} = J_0 \delta x_k + J_1 \delta x_{k-d} + J_m \delta M_k.$$

The stability of x^ is determined by the eigenvalues of the lifted companion Jacobian. Parameter sweeps that increase response gain, delayed coupling, or memory persistence move these eigenvalues toward or beyond the unit circle.*

Proof. Linearization of the lifted map gives an ordinary finite-dimensional map on history and memory coordinates. Local discrete-time stability is equivalent to all eigenvalues of the companion Jacobian lying inside the unit disk. Increasing the norms of coupling and persistence terms changes this Jacobian continuously; instability occurs when eigenvalues cross the unit circle. $\square$

4.1 OmegaHive1 order parameters

For diagnosis, define low-dimensional order parameters:

$$R_k = \left| \frac{1}{N} \sum_{j=1}^N e^{i\phi_{j,k}} \right| \quad (\text{coordination/synchrony}),$$

$$H_k = - \sum_a p_k(a) \log p_k(a) \quad (\text{behavioral or semantic diversity}),$$

$$C_k = x_k^\top L x_k \quad (\text{network tension across graph Laplacian } L),$$

$$Q_k = \text{artifact-progress or quality metric}, \quad B_k = \text{resource/backlog pressure}.$$

Analyze trajectories in (R, H, C, Q, B) as well as in raw state space.

Design Principle 1 (Do not search for chaos in raw logs first). *Raw agent traces are too high-dimensional and semantically mixed. First define scrutable observables and order parameters, then test whether the induced observable dynamics has bounded recurrence, sensitivity, and nonperiodicity.*

5 S4. Theory-motivated OmegaSim instructions

5.1 Implementation targets

1. Implement the lifted delayed state $z_k = (x_k, \dots, x_{k-m}, M_k)$ explicitly. Do not hide delays in ad hoc buffers without logging them.
2. Define a compact state budget. Use saturating nonlinearities, normalization, or explicit resource conservation so runs cannot be mistaken for attractors while diverging.
3. Expose the control ratios $\rho, \delta, \mu, \kappa, \nu$ as primary sweep parameters. Raw constants should be derived from these ratios where possible.
4. Compute the fixed point or empirical baseline regime at low gain. Use it as the contraction/control case.
5. Linearize the deterministic skeleton around equilibria or recurrent regimes. Track eigenvalues of the lifted companion Jacobian as parameters change.
6. Run sweeps that cross from $\rho < 1$ to $\rho > 1$, from $\delta \approx 0$ to $\delta = O(1)$, and from weak to moderate memory persistence μ .
7. Near transitions, run hysteresis sweeps upward and downward in gain/delay to detect multistability and crises.
8. Estimate finite-time Lyapunov exponents with paired trajectories, periodic renormalization, multiple perturbation directions, and matched noise paths.
9. Add null controls: no-delay, linearized nonlinearity, degree-preserving rewiring, delay-shuffled, phase-randomized surrogate, and frozen-noise paired comparisons.
10. Classify each region as fixed point, limit cycle, quasiperiodic, candidate chaotic, transient chaotic, noise-driven irregular, divergent, or trivial saturation.

5.2 Specific suggested sweep grid

A first OmegaSim phase diagram should sweep:

$$\rho \in [0.2, 3.0], \quad \delta \in \{0, 0.1, 0.3, 1, 3\},$$
$$\mu \in \{0.5, 1, 3, 10\}, \quad \kappa \in \{0, 0.05, 0.2, 0.8\}, \quad \nu \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}.$$

The exact ranges should be rescaled after measuring the relaxation time and state norms of the implemented simulator.

5.3 Acceptance criteria for a candidate strange-attractor region

A region should only be reported as a candidate complex/strange-attractor regime if it passes all of these gates:

1. boundedness gate: norms and resource variables remain in a nontrivial compact range;
2. refinement gate: the qualitative regime survives smaller timestep or longer history resolution;
3. perturbation gate: positive finite-time Lyapunov estimate under fixed noise path;
4. surrogate gate: behavior differs from no-delay, linearized, and phase-randomized controls;
5. semantic gate: Hyperseed event records identify plausible feedback loops generating the dynamics;
6. task gate: the regime is related to useful simulation behavior, not just aesthetic irregularity.

6 Conclusion

The rigorous core is modest but useful. If the hive update is ill-posed, unbounded, or contractive, complex attractor claims are unsupported or impossible. If the system is well-posed, bounded, non-contractive, delayed, and nonlinear, then gain, delay, memory persistence, modular leakage, and noise become principled coordinates for exploring bifurcations and candidate strange-attractor regimes.

The practical OmegaSim instruction is therefore: build the simulator so these coordinates are explicit; enforce boundedness; sweep through the contraction boundary; test delay-induced instability; and only call a regime strange-attractor-like after Lyapunov, recurrence, surrogate, refinement, and semantic-provenance checks pass.

Revision Addendum (2026-07-01): Cognitive Appraisal, Residual Lobes, and New Experiment Families

This addendum responds to the 2026-07-01 feedback on the original note. It preserves all prior material and adds eight substantive extensions.

R1. Logistic Response as Bounded Thresholded Cognitive Appraisal

The original note used a generic saturating nonlinearity S . We now interpret the nonlinear response as a *cognitive appraisal function*: each agent or hive component evaluates incoming signals through a bounded, thresholded transform—not as arbitrary chaos injection, but as a principled encoding of attention allocation, urgency, and resource commitment.

Definition 8 (Cognitive appraisal response). *The componentwise response is a logistic (sigmoid) function*

$$\sigma_\alpha(x; \theta) = \frac{1}{1 + \exp(-\alpha(x - \theta))},$$

where $\alpha > 0$ is the appraisal steepness (sensitivity to deviations from threshold θ). The amplitude $\beta > 0$ is applied outside the unit sigmoid as the saturation level (asymptotic maximum response). The effective response in the hive update replaces $S(\cdot)$ with

$$S_i(\cdot) \mapsto \beta_i \sigma_{\alpha_i}(w_i^\top x_k + b_i; \theta_i),$$

giving each component i its own threshold, steepness, and amplitude.

Observation 1 (Appraisal threshold as bifurcation coordinate). *The appraisal steepness α acts as a gain parameter: sufficiently low α suppresses sensitivity to deviations from threshold, while high α creates a switching-like regime with sharp transitions. The threshold θ shifts the operating point. Together (α, θ) define a natural two-parameter sweep from low-gain response to thresholded multi-state dynamics. Whether an actual bifurcation occurs depends on the surrounding hive update map and must be checked numerically in each experiment.*

Justification 1. *For $\alpha \rightarrow 0$, the unit sigmoid approaches the constant $1/2$ and its maximum derivative tends to zero. For finite α , the response $\beta\sigma_\alpha$ has maximum derivative $\alpha\beta/4$ at the threshold. For $\alpha \rightarrow \infty$, the response approaches a scaled Heaviside step. Thus $\alpha\beta/4$ is a local gain bound for the appraisal nonlinearity, not by itself a proof of global contraction or bifurcation.*

R2. Latent Motivational State with Explicit Dynamics

Definition 9 (Latent motivational state). *Introduce a latent motivational vector $\mu_k \in \mathbb{R}^p$ with explicit dynamics*

$$\mu_{k+1} = (1 - \lambda)\mu_k + \lambda \Phi(x_k, \mu_k) + \zeta_k,$$

where $\lambda \in (0, 1)$ is the motivation update rate, Φ is a feedback function from current hive state and current motivation to updated motivation, and ζ_k is endogenous drive noise. The motivation μ_k modulates the appraisal thresholds:

$$\theta_i(\mu_k) = \theta_i^{(0)} + \gamma_i^\top \mu_k,$$

so that motivational state shifts what deviations are attended to.

Assumption 1 (Motivational timescale separation). *We assume $\lambda \ll 1/T_{\text{decision}}$, i.e., motivational state evolves slower than the decision update. This yields a slow-fast system where x_k is the fast variable and μ_k is the slow variable, enabling singular-perturbation analysis.*

Proposition 5 (Slow motivation induces bifurcation sweeping). *As μ_k evolves slowly, the appraisal thresholds $\theta_i(\mu_k)$ drift, causing the fast dynamics' bifurcation parameters to sweep. This can produce intermittent visits to multiple attractor lobes, hysteresis across bifurcation boundaries, and crisis-induced switching—all driven by internal motivational state rather than external forcing.*

R3. Separate Semantic and Artifact Fields and Their Coupling

Definition 10 (Semantic field). *The semantic field $s_k \in \mathbb{R}^{d_s}$ captures the representational content flowing through the hive: message semantics, task descriptions, shared memory content, and interpretive type assignments. Its dynamics are*

$$s_{k+1} = \Psi_s(s_k, x_k, \mu_k) + \xi_k^s,$$

where Ψ_s may involve embedding-space operations, semantic similarity weighting, and type-dependent routing.

Definition 11 (Artifact field). *The artifact field $a_k \in \mathbb{R}^{d_a}$ tracks produced artifacts: code, documents, test results, structured outputs. Its dynamics are*

$$a_{k+1} = \Psi_a(a_k, s_k, x_k, \mu_k) + \xi_k^a,$$

where Ψ_a includes quality evaluation, version tracking, and resource cost accounting.

Definition 12 (Field coupling). *The coupled system replaces the monolithic state x_k with*

$$\tilde{x}_k = (x_k^{\text{op}}, s_k, a_k, \mu_k),$$

where x_k^{op} are operational/coordination variables. The semantic field influences operational dynamics through appraisal thresholds $\theta_i(s_k)$, and the artifact field feeds back through resource load $r_k = g(a_k)$ and quality metrics $q_k = Q(a_k)$. Bidirectional coupling strength is parameterized by χ_{sa} (semantic→artifact) and χ_{as} (artifact→semantic).

Design Principle 2 (Separate fields enable distinct diagnostics). *Treating s_k and a_k as distinct dynamical fields allows separate Lyapunov analysis, attractor characterization, and surrogate controls for each. Semantic complexity without artifact complexity, or vice versa, is diagnostically distinct from joint complexity.*

R4. Costly Prediction and Resource Accounting

Definition 13 (Prediction cost). *Each predictive step—whether by an agent predicting outcomes or by the hive forecasting its own state—incurs a resource cost*

$$c_k^{\text{pred}} = c_{\text{fixed}} + c_{\text{var}} \|p_k\|^2,$$

where p_k is the prediction depth or model complexity and $c_{\text{var}} > 0$. Total resource consumption must satisfy

$$\sum_i c_{k,i}^{\text{pred}} + c_{k,i}^{\text{act}} \leq R_{\text{max}},$$

where R_{max} is the hive-wide resource budget per timestep.

Definition 14 (Resource-constrained dynamics). *The hive update with resource accounting becomes*

$$x_{k+1} = S(W_0 x_k + W_1 x_{k-d} + W_m M_k + B u_k + \xi_k; R_{\text{max}} - c_k^{\text{pred}}),$$

where the remaining budget $R_{\text{max}} - c_k^{\text{pred}}$ gates the effective amplitude of activation. When prediction costs are high, effective gain drops, pushing the system toward the contractive regime; when prediction is cheap, full nonlinear dynamics operate.

Design Principle 3 (Cost-induced self-regulation). *If prediction cost is chosen to increase with a measured complexity proxy, such as a local finite-time Lyapunov estimate or forecast-error budget, then highly chaotic regimes incur higher costs, reducing effective gain and creating a self-regulating feedback loop. This is a proposed design mechanism for bounded complexity; the strength and stability of the mechanism must be measured in the A7 experiments rather than assumed analytically.*

R5. Hysteresis and Adaptive Thresholds

Definition 15 (Adaptive appraisal thresholds). *Appraisal thresholds evolve according to*

$$\theta_{i,k+1} = \theta_{i,k} + \eta_i (\sigma_{\alpha_i}(w_i^\top x_k; \theta_{i,k}) - p_i^*),$$

where $p_i^* \in (0, 1)$ is a target activation rate and $\eta_i > 0$ is the adaptation rate. This drives each component toward a target operating point, implementing homeostatic regulation.

Hypothesis 1 (Adaptive thresholds can create hysteresis). *When the adaptation rate η_i is small but nonzero, the threshold $\theta_{i,k}$ tracks the recent activation level with a finite relaxation time. On upward parameter sweeps (increasing gain α), the threshold may rise to suppress over-activation; on downward sweeps, it may lower only after a delay. This lag can produce hysteresis: the system’s state at gain α on the upward sweep may differ from its state at the same α on the downward sweep. The loop width should be reported empirically as a function of the adaptation rate η_i and the imposed sweep rate ϵ .*

Justification 2. *Let $\bar{\theta}(\alpha)$ denote a quasi-steady threshold, when such a branch is well defined. Under a sweep $\alpha_k = \alpha_0 + \epsilon k$, the adaptive update is a first-order lag rather than instantaneous equilibration. The sign of the lag changes with sweep direction, which is the qualitative source of the hysteresis test. A universal scaling law such as η_i/ϵ is not asserted here; the relevant scaling depends on the chosen update convention and will be estimated in the proposed sweep.*

R6. Residual-State Lobe Discovery

Definition 16 (Residual state). *Define the residual state as the deviation of the full system state from the best-fit low-dimensional attractor model:*

$$\rho_k = \tilde{x}_k - \Pi_{\mathcal{A}}(\tilde{x}_k),$$

where $\tilde{x}_k = (x_k^{\text{op}}, s_k, a_k, \mu_k)$ is the full coupled state, $\Pi_{\mathcal{A}}$ is the projection onto the fitted attractor model $\mathcal{A}$ (e.g., a principal manifold, delay-coordinate embedding, or low-order DMD approximation), and ρ_k is the residual. The residual captures structure not explained by the low-dimensional attractor model.

Definition 17 (Residual lobe). *A residual lobe is a persistent cluster in residual-state space $\{\rho_k\}$ identified by a clustering algorithm (e.g., Gaussian mixture, HDBSCAN, or k-medoids in delay-coordinate space) with the following properties:*

1. Persistence: *the cluster survives across multiple validation windows of length $\geq 10 T_{\text{relax}}$.*
2. Separation: *the cluster’s centroid is at least 2σ from other cluster centroids in residual norm.*
3. Reproducibility: *the cluster appears in $\geq 80\%$ of independent runs with different initial histories under the same parameter setting.*
4. Semantic distinguishability: *Hyperseed event records for trajectories within the lobe differ systematically (in agent roles, artifact types, or semantic content) from records for other lobes.*

Design Principle 4 (Lobe discovery as a validation tool). *Residual lobe discovery provides a concrete method for detecting whether the full system has multi-lobe attractor structure that is invisible to the low-dimensional observable. If lobes are semantically distinguishable, they identify distinct cognitive regimes (e.g., exploration vs. exploitation, problem-attacking vs. reflective) that the hive transitions between.*

R7. Amplitude-Matched Linear Controls

Definition 18 (Amplitude-matched linear control). *Given a nonlinear hive system $x_{k+1} = G(x_k, \dots, x_{k-m})$ with operational state norm $\|x_k\| \leq X_{\max}$ in the regime of interest, the amplitude-matched linear control is*

$$x_{k+1}^{\text{lin}} = A_0 x_k + A_1 x_{k-d} + B u_k + \bar{c},$$

where A_0, A_1 are obtained by linearizing G around the regime mean $\bar{x} = \frac{1}{T} \sum_k x_k$ and $\bar{c}$ is chosen so that $\|x_k^{\text{lin}}\| \approx \|x_k\|$ in the matched regime. The linear control uses the same noise path ξ_k (common random numbers) and the same delay structure as the nonlinear system.

Proposition 6 (Linear control as a surrogate baseline). *The amplitude-matched linear control isolates the contribution of nonlinearity from delay and noise. If the linear control exhibits comparable spectral breadth, Lyapunov exponent, or recurrence statistics, the nonlinear system’s complexity is attributable to delay/noise, not to nonlinear structure. Only when the nonlinear system significantly exceeds the linear control on these metrics can nonlinear strange-attractor claims be supported.*

Proof. The linear control shares delay structure, noise realization, and amplitude with the nonlinear system; it differs only in the replacement of G by its linearization. Therefore any excess complexity in the nonlinear system is attributable to the nonlinear terms. This is a standard matched-control argument from the design-of-experiments tradition. $\square$

R8. Experiment Families A6, A7, and A8

A6: Cognitive Appraisal Bifurcation Scan

Hypothesis: Varying appraisal steepness α and threshold θ across the hive produces a smooth bifurcation from contractive fixed-point dynamics through damped oscillation to multi-lobe recurrent dynamics, mediated by the cognitive appraisal function rather than by external noise injection.

Sweep variables:

- Appraisal steepness $\alpha \in \{0.5, 1, 2, 5, 10, 20\}$
- Threshold $\theta \in \{-2, -1, 0, 1, 2\}$ (relative to baseline input distribution)
- Motivational coupling $\gamma \in \{0, 0.1, 0.5, 1.0\}$

Controls:

- Amplitude-matched linear control (R7) at each (α, θ) point
- No-motivational-dynamics control ($\gamma = 0$, frozen μ_k)
- Phase-randomized surrogate

Acceptance criteria:

1. Bifurcation diagram in (α, θ) shows a qualitative transition from fixed-point to oscillatory to multi-lobe dynamics.
2. The transition region has positive finite-time Lyapunov exponent while the linear control does not.
3. Motivational coupling ($\gamma > 0$) widens the multi-lobe region compared to $\gamma = 0$.
4. Hysteresis is observable on upward vs. downward α sweeps with adaptive thresholds.

A7: Residual Lobe Discovery and Validation

Hypothesis: The full coupled system $(x^{\text{op}}, s, a, \mu)$ has multi-lobe attractor structure that is partially invisible to the low-dimensional observable $O(E_{\leq t})$. Residual lobes are semantically distinguishable.

Sweep variables:

- Attractor model dimension $d_{\mathcal{A}} \in \{2, 3, 5, 8\}$
- Clustering algorithm: HDBSCAN, GMM, k -medoids
- Delay embedding dimension for residual space: $m_{\text{emb}} \in \{3, 5, 10\}$
- Validation window length: $\{5, 10, 20\} \cdot T_{\text{relax}}$
- Number of independent runs: ≥ 20 per parameter setting

Controls:

- Random-label surrogate: cluster assignments randomly permuted across time to test whether semantic distinguishability exceeds chance.
- Single-field controls: residual computed using only x^{op} , only s , only a , or only μ , to test which field’s residual carries lobe structure.
- Null-model residual: residual computed against a trivial (zero-order) attractor model, i.e., the mean state, to verify that the clustering captures genuine attractor structure and not just noise.

Acceptance criteria:

1. At least one residual lobe satisfies all four properties from the residual lobe definition (R6): persistence, separation, reproducibility, and semantic distinguishability.
2. The random-label surrogate produces $< 5\%$ false-positive lobe rate at the same clustering thresholds.
3. Single-field controls identify which field(s) contribute most to lobe structure, enabling targeted interpretation.
4. Lobe membership predicts subsequent behavioral regime with above-chance accuracy ($p < 0.01$ by permutation test).

A8: Costly Prediction and Self-Regulation Experiment

Hypothesis: When prediction cost scales with dynamical complexity (high local Lyapunov exponent $\Rightarrow$ higher c_k^{pred}), the hive system self-regulates into a bounded-complexity regime: chaotic episodes are automatically damped by resource depletion, and the system settles into an intermittent regime alternating between high-complexity bursts and resource-recovery quiescence.

Sweep variables:

- Prediction cost coefficient $c_{\text{var}} \in \{0, 0.01, 0.1, 0.5, 1.0\}$
- Resource budget $R_{\text{max}} \in \{0.5, 1, 2, 5\}$ (relative to median unconstrained activation cost)

- Cost complexity coupling: $c_k^{\text{pred}} \propto |\lambda_{\text{FTLE}}|^p$ with exponent $p \in \{0, 1, 2\}$ (no coupling, linear, quadratic)
- Prediction depth $p_k \in \{1, 2, 5, 10\}$ steps ahead

Controls:

- Zero-cost control ($c_{\text{var}} = 0$): standard dynamics without resource accounting, to establish the unconstrained complexity baseline.
- Fixed-cost control: $c_k^{\text{pred}} = c_{\text{fixed}}$ (constant regardless of complexity), to distinguish resource scarcity effects from complexity-coupled self-regulation.
- Amplitude-matched linear control (R7) with the same resource budget, to separate nonlinear self-regulation from linear resource gating.

Acceptance criteria:

1. With cost-complexity coupling ($p > 0$), the system exhibits intermittent complexity: episodes of positive finite-time Lyapunov exponent alternated with quiescent recovery periods.
2. The zero-cost control shows sustained chaos (no self-damping), confirming that the intermittent pattern is caused by cost coupling, not by other parameters.
3. The fixed-cost control shows uniform suppression (no intermittency), confirming that the intermittency requires complexity-dependent cost, not just resource scarcity.
4. Long-run complexity (time-averaged Lyapunov exponent) decreases monotonically with c_{var} for fixed R_{max} , demonstrating a tunable self-regulation knob.
5. Hysteresis in the complexity–cost relationship is observable when c_{var} is swept up and down, due to adaptive thresholds (R5).

Summary of revisions

This addendum adds: (R1) cognitive appraisal interpretation of the logistic response, (R2) latent motivational state with slow–fast dynamics and threshold modulation, (R3) separate semantic and artifact fields with bidirectional coupling, (R4) costly prediction with resource accounting and self-regulating complexity, (R5) adaptive thresholds producing hysteresis, (R6) residual-state lobe discovery with clustering and validation, (R7) amplitude-matched linear controls isolating nonlinear contributions, and (R8) three new experiment families (A6, A7, A8) with explicit hypotheses, sweep variables, controls, and acceptance criteria. All original material is preserved; this content is appended via

input before
enddocument.

Replication Status (2026-07-16): Bounded Proxy Result

This subsection records a measured result only. It does not strengthen the strange-attractor claim or the semantic-grammar (Hyperseed feedback-loop) claim of this note. Those claims remain conjectural and gated by the acceptance criteria stated above; none of the six candidate-regime gates (boundedness, refinement, perturbation/FTLE, surrogate, semantic, task) is asserted as passed on the basis of the result below.

Observed. A repaired untouched-seed proxy replication (OmegaSim frozen-CLA restart; proxy run pinned to OmegaSim 18c7408 and chaoslang 974af31) reproduced the specific parameter cell (coupling = 0.60, delay = 3, roles = 8) in 4/5 untouched seeds — i.e. four of five seeds yielded that cell. This 4/5 figure and the earlier 1/12 figure have *different denominators* and are not the same quantity: 1/12 is one promoted cell among twelve tested cells (a cell-selection rate), whereas 4/5 is four seed-wins among five seeds *for that already selected cell* (a per-cell seed-reproduction rate). The 4/5 figure therefore does *not* supersede the 1/12 figure; both stand, at different levels of the pipeline.

Epistemic status. *Bounded proxy evidence; replication complete for the proxy, formal claims blocked.* The 4/5 seed-reproduction rate was obtained under exact controls (matched controls were present, not absent); replication of the bounded proxy result is complete. Nevertheless the strange-attractor and semantic-grammar (Hyperseed feedback-loop) claims remain *blocked*, because the external Mackey–Glass / Lorenz–96 calibration *failed at the coding stage* and so the six candidate-regime gates (boundedness, refinement, perturbation/FTLE, surrogate, semantic, task) cannot be adjudicated against an external chaotic baseline. This datum is thus a completed, exact-controlled proxy replication, but not confirmation of a strange-attractor regime and not evidence for a Hyperseed semantic grammar; formal promotion awaits a *working* external calibration harness.

Provenance. Recorded per Ben’s instruction of 2026-07-16 to revise Note 0004 *only* to record the bounded proxy result, leaving the attractor and semantic-grammar claims unstrengthened. Proxy lineage pins: OmegaSim 18c7408, chaoslang 974af31; cell (0.60, 3, 8) reproduced in 4/5 untouched seeds under exact controls; cell-selection rate 1/12 (distinct denominator). Note: c8e9e91 is a *separate, stricter held-out calibration path* and is *not* the lineage of this proxy run. Formal claims blocked pending a working external Mackey–Glass / Lorenz–96 calibration (prior attempt failed coding).